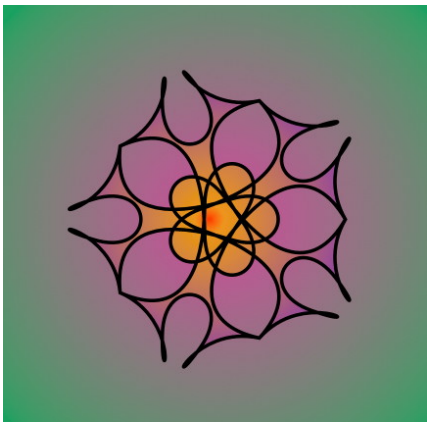


Group Theory

An Introduction to Algebraic Symmetry and Structures

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1 Introduction to Groups

1.1 Basic Notions

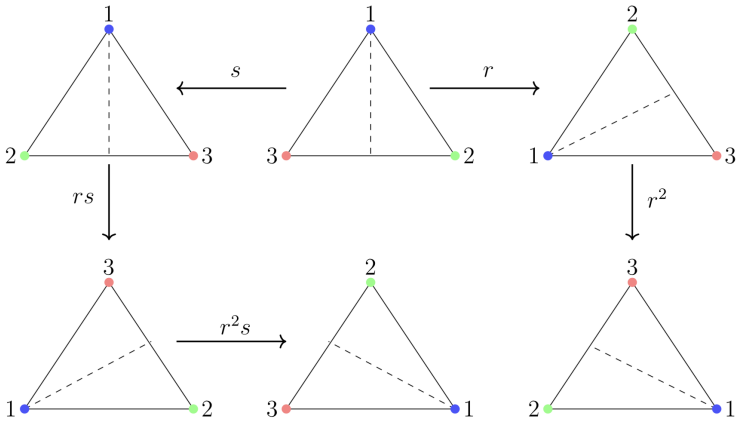
Definition 1.1 (Group)**.** Let G be a set, and we define a binary operation $* : G \times G \to G$, then $(G, *)$ is said to be a group if it follows the following axioms:

1. $\forall\, a, b \in G \implies a * b \in G$, and we say the set has **closure**.
2. $\forall\, a, b, c \in G$ we have $a * (b * c) = (a * b) * c = a * b * c$, so the operation is **associative**.
3. $\exists\, e \in G \,\forall\, a \in G$ such that $a * e = e * a = a$, where e is called the **identity** of the group.
4. $\exists\, a^{-1} \in G \,\forall\, a \in G$ such that $a * a^{-1} = a^{-1} * a = e$, where a^{-1} is called the **inverse** of an element a.

That's a particularly heavy seeming definition, but it happens to be incredibly simple, and we show its simplicity by listing a number of examples:

Example 1.2.

1. Consider an equilateral triangle, lets label each corner with a colour. Now we perform operations on this triangle that change the order of the colours, which are *rotations* (r) and *reflections* (s).



   If we consider these *set* of operations to be elements, and take the binary operation to be performing one operation after another, then we have a group! This is generally called the dihedral group D_{2n}, where n is the number of sides the regular n–gon has. It is the group of symmetries that a regular n–gon possesses, n rotations and n reflections. The action of doing nothing is also an element of this group, as every group must have an identity.

2. The natural numbers $\mathbb{N}$, integers $\mathbb{Z}$, rational numbers $\mathbb{Q}$, real numbers $\mathbb{R}$ and complex number $\mathbb{C}$ are groups on the operation of addition.

3. On multiplication, only $\mathbb{Q}^*, \mathbb{R}^*$ and $\mathbb{C}^*$ are groups, since elements of $\mathbb{Z}^*$ do not have multiplicative inverses (reciprocals) lying in $\mathbb{Z}^*$. Here, $F^* = F - \{0\}$, since the inverse of zero is undefined in all sets.

4. Vector spaces on real or complex fields are also groups by defintion.

5. The set of $n \times n$ non–singular matrices with entries from a set X form a group upon matrix multiplication, and this is called the general linear group.

$$\mathrm{GL}(n, X) = \left\{A = (a_{ij})_{n\times n} \mid a_{ij} \in X, \det(A) \neq 0\right\} \tag{1}$$

6. All the valid twisting moves of a Rubik's Cube combined in sequence form a finite group.

7. Let us take the days of the week, and assign a number to each of them with Sunday being 0 and Saturday being 6. So we have a set $\{0, 1, 2, 3, 4, 5, 6\}$, now we define an operation where we add two days together, say $3 + 6 = 9$, but the day 6 days after Thursday is Wednesday, which is labelled by 2. So in general, the operation is addition modulo 7 (modulo 7 just means remainder after division by 7). We have an identity element 0, and the inverse for an element k would be $7 - k$. Addition is associative and the addition will will give us an element in the set. We have formed a group! In general this set, $\mathbb{Z}_n = \{0, 1, 2, \ldots n - 1\}$ is called the group of residue classes modulo n.

8. A similar group to $\mathbb{Z}_n$ is U_n, which contains elements which are coprime to n i.e. integers k with $\gcd(n, k) = 1$, with the operation being multiplication modulo n. For example: $U_{12} = \{1, 5, 7, 11\}$, with the inverse k^{-1} of an element k being the solution to $k^{-1}k = 1 \pmod{n}$.

9. The set of the complex solutions to $x^n = 1$, called the nth roots of unity also form a group under complex multiplication.

$$\mathbb{C}_n = \left\{e^{i2k\pi/n} \mid 0 \leq k < n\right\} \tag{2}$$

Now, we shall prove some basic lemmas of groups that we usually take for granted. From here on, we omit the $(*)$ when mentioning whether something is a group, and when taking the operation between elements.

Lemma 1.3.

1. **(Left Cancellation)** $\forall\, a, b, c \in G$ *such that* $ab = ac \implies b = c$.
2. **(Right Cancellation)** $\forall\, a, b, c \in G$ *such that* $ba = ca \implies b = c$.
3. *There is only one unique identity element for any group.*
4. *There is only one unique inverse for an element in a group.*

Proof.

1. Given that $ab = ac$, for some $a, b, c \in G$, by the existence of inverse, we write $a^{-1}(ab) = a^{-1}(ac)$ which turns into $a^{-1}ab = a^{-1}ac$ by associativity and into $eb = ec$ then we have $b = c$.

□

2. Given that $ba = ca$, for some $a, b, c \in G$, by the existence of inverse, we write $(ba)a^{-1} = (ca)a^{-1}$ which turns into $baa^{-1} = caa^{-1}$ by associativity and into $be = ce$ then we have $b = c$.

□

3. Let e' and e be two identities of a group G. Then, $ae = ea = a$ and $ae' = e'a = a \ \forall \ a \in G$. So we have $a = ae = ae'$ and by left cancellation we get $e = e'$, hence the two identities are one and the same.

□

4. Let a' and a'' be two inverses of an element $a \in G$, then $aa' = a'a = e$ and $aa'' = a''a = e$. So we have $e = a'a = a''a$ and by right cancellation we have $a' = a''$, hence the two inverses are one and the same.

□

Definition 1.4 (Abelian Group)**.** A group G is said to be an abelian group if and only if $\forall \ a, b \in G$, we have $ab = ba$, i.e. the operation is commutative on G. Hence, it is typically also called a commutative group. It was named after Niels Henrik Abel, one of the founders of group theory.

Definition 1.5 (Subgroup)**.** Let G be a group, and we have a set $H \subset G$. Then, H is called a subgroup of G if and only if H is also a group, denoted by $H < G$.

Example 1.6.

1. Aside from the obvious ones like $\mathbb{Q} < \mathbb{R}$, lets look at the group $\mathrm{SL}(n, X)$, the special linear group which consists of non singular $n \times n$ matrices with unit determinant. This is a subgroup of the general linear group $\mathrm{GL}(n, X)$.

$$\mathrm{SL}(n, X) = \left\{A = (a_{ij})_{n\times n} \mid a_{ij} \in X, \det(A) = 1\right\} < \mathrm{GL}(n, X) \tag{3}$$

2. The group of integer multiples of an integer a as $\mathbb{Z}a$, which is a subgroup of $\mathbb{Z}$.

$$\mathbb{Z}a = \{n \in \mathbb{Z} \mid n = ka \text{ for some } k \in \mathbb{Z}\} \tag{4}$$

Proposition 1.7 (Subgroup Criteria)**.**
i. *If G is group and $H \subset G$, then H is a subgroup if and only if $\forall \ a, b \in H \implies ab^{-1} \in H$.*

ii. *If G is a group and $H \subset G$, then H is a subgroup if and only if $\forall \ a, b \in H \implies ab \in H \quad$ and $a^{-1} \in H$.*

Proof.
i. ($\implies$) If H is a subgroup, then by closure and existence of inverse, $ab^{-1} \in H$ for every $a, b \in H$.
($\impliedby$) If $\forall \ a, b \in H \implies ab^{-1} \in H$, then $aa^{-1} = e \in H$, which means that $\forall \ a \in H$ we have $ea^{-1} = a^{-1} \in H$, and $\forall \ a, b^{-1} \in H$, we have $ab \in H$. This means we have existence of identity, inverse and closure, so H is group and since it is a subset of G, thus $H < G$.

□

ii. Both sides are directly implied by the group axioms. Thus, $H < G$. □

Definition 1.8 (Order)**.** If we have a group G with a finite number of elements, then that number of elements is the order of the group, written as $|G|$. If the group is an infinite group, then $|G| = \infty$.
The order of an element $a \in G$ is the number of times it must be composed with itself to get the identity i.e. if $a^k = e$, then the order of a, written as $\mathcal{O}(a) = k$. If $a^k = e$ never occurs for any k, then $\mathcal{O}(a) = \infty$.

Lemma 1.9. *If G is a group, and we have $\mathcal{O}(a) = n$ for some $a \in G$, then if we have $a^m = e \implies n \mid m$ (n divides m).*

Proof. By the definition of order of an element, n is the smallest integer s.t. $a^n = e$. It follows that $(a^n)^k = a^{nk} = e$, and only multiples of n can give us an identity, Hence, m must also be a multiple of n and thus $n \mid m$. □

Lemma 1.10. *Let G be a group, and let $a \in G$ with $\mathcal{O}(a) = n$. Then $\mathcal{O}(a^k) = n/\gcd(n,k)$.*

Proof. Let $\mathcal{O}(a^k) = p$, and by definition p is the least integer such that $(a^k)^p = a^{kp} = e$. Then, by the previous lemma we know that $n \mid kp \implies \frac{n}{\gcd(n,k)} \mid \frac{kp}{\gcd(n,k)}$, and since $n/\gcd(n,k)$ and $k/\gcd(n,k)$ share no common factors now, then $\frac{n}{\gcd(n,k)} \mid p$. Therefore, what we have is $(a^k)^p = e \iff \frac{n}{\gcd(n,k)} \mid p$, and once again by the previous lemma, the order of a^k is $\mathcal{O}(a^k) = n/\gcd(n,k)$. □

Definition 1.11 (Generating Set)**.** If we have some elements of a group G whose powers or inverses one can compose to form all the elements of the group, then the set of those elements is called a generating set. If the set S generates G, we write $G = \langle S \rangle$.

Example 1.12. The set of integers is completely generated by $\{1\}$, since we can add it to itself to produce all the positive integers, and its inverse -1 for all the negative integers. The same is true for $\mathbb{Z}_n$ being generated by $\{1\}$ and $\mathbb{C}_n$ being generated by $\{e^{2i\pi/n}\}$. In fact, $\mathbb{Z}_n$ and $\mathbb{C}_n$ have a very important relation between that we shall explore later.

Definition 1.13 (Cyclic Group)**.** A group G is said to be cyclic, if $\exists\, a \in G$ such that $G = \langle \{a\} \rangle$, i.e. the entire group is generated by a single element. If $|G| = n$, then

$$G = \left\{e, a, a^2, \dots a^{n-1}\right\}$$

If G is of infinite order, then:

$$G = \left\{e, a^{\pm 1}, a^{\pm 2}, \dots\right\}$$

Proposition 1.14. *Let $G = \langle a \rangle$ be a cyclic group, then:*

1. *Every subgroup of G is cyclic.*
2. *If G is of infinite order, then every subgroup of G is of the form $H = \left\langle a^k \right\rangle$ where $k \in \mathbb{Z} - \{0\}$.*
3. *If $\mathcal{O}(a) = n$, then every subgroup of G is of the form $H = \left\langle a^{n/p} \right\rangle$ where $p \mid n$.*

Proof.

1. Given that $G = \langle a \rangle$, let H be a proper subgroup of G. Now, let m be the least integer such that $a^m \in H$, then by closure and existence of inverse, any element of the form $(a^m)^k = a^{km} \in H$, where $k \in \mathbb{Z} - \{0\}$. Now, if we assume $a^{km+1} \in H$, then $(a^{km+1})^p = a^{pkm}a^p \in H \implies a^p \in H$ for any $p \in \mathbb{Z} - \{0\}$. But then it would cease to be a proper subgroup of G since $a^p \in H \implies H = G$. Therefore only elements of the form $(a^m)^k$ are in the the group, hence $H = \langle a^m \rangle$ is a cyclic group.

□

2. It follows as a corollary of the above proof that every subgroup of G can be written as $H = \left\langle a^k \right\rangle$ where $k \in \mathbb{Z} - \{0\}$.

□

3. Given that $\mathcal{O}(a) = n$ then let k be least integer such that $a^k \in H$ and $\left\langle a^k \right\rangle = H$. Now, using the division algorithm we write $n = pk + r \implies e = a^n = a^{qk}a^r$, then $a^r = a^{-pk} \in H$. We chose k to be smallest integer such that a^k generates H, but $r < k$, so the only way $a^r \in h$ is for $r = 0$, so we then get $k = n/p$ and therefore every subset of G is of the form $\left\langle a^{n/p} \right\rangle$, where $p \mid n$.

□

1.2 Equivalence and Cosets

Definition 1.15 (Equivalence)**.** Two elements $a, b \in S$ are said to be *equivalent*, written as $a \sim b$, if there exists an equivalence relation on the set S. An equivalence relation is required to be follow the given rules:

- **Transitivity**: If $a \sim b$ and $b \sim c$, then $a \sim c$.
- **Symmetric**: If $a \sim b$, then $b \sim a$.
- **Reflexive**: $\forall\, a \in S$, we have $a \sim a$.

Example 1.16.

1. Defining an equivalence relation $a \sim b \iff 2 \mid a - b$ for $a, b \in \mathbb{Z}$, then all even numbers are equivalent to each other, and all odd number are equivalent to each other. This equivalence relation essentially partitions $\mathbb{Z}$ into a set of odd and even integers.

2. Let S be the set of all triangles in a plane. If $a, b \in S$, define $a \sim b$ if a and b are similar if a and b have corresponding angles that are the same.

3. Let S be the set of all polynomials with real coefficients. If $f, g \in S$, define $f \sim g \iff f' = g'$, where f' is the derivative of f. Then $\sim$ is an equivalence relation on S.

Definition 1.17 (Equivalence Class)**.** If $\sim$ is an equivalence relation on a set S and $a \in S$, then the equivalence class of a is defined as $[a] = \{x \in S \mid x \sim a\}$.

Proposition 1.18. *Given an equivalence relation $\sim$ on a set S, it then determines a partition on S. Conversely, for any partition P of S, there is an equivalence relation on S whose equivalence classes are the elements of P.*

Proof. Let $[a]$ and $[b]$ be two equivalence classes for distinct elements $a, b \in S$. Assuming $[a] \cap [b] \neq \emptyset$, then for some $x \in [a] \iff x \in [b]$, so $x \sim a$ and $x \sim b$. But by transitivity, $a \sim b$ and therefore $[a] = [b]$. This is true for any set S if and only if $a = b$, but this contradicts our assumption that a and b are distinct. Therefore, an equivalence relation partitions a set into pairwise disjoint subsets.

To prove the converse, let P be a collection of nonempty pairwise disjoint subsets of S, whose union is S. Then, we define an equivalence relation $a \sim b$ for $a, b \in S$ if both a and b belong to the same subset in the collection P. Then, we have the partition determining the equivalence relation. □

Definition 1.19 (Coset)**.** Let G be a group and $H < G$ is a subgroup of G. If we have a fixed element $a \in G$, then the *left coset* of H in G is defined as:

$$aH = \{ah \mid h \in H\} \tag{5}$$

and the *right coset* of H in G is defined to be:

$$Ha = \{ha \mid h \in H\} \tag{6}$$

Cosets may seem like a very specific defintion that does not seem particularly motivated, but they happen to one of the most useful tools in group theory.
From here on, unless mentioned, coset refers specifically to the left coset.

Lemma 1.20. *Let G be a group and $H \leq G$. Then for $x, y \in G$, the following are equivalent:*

1. $xH = yH$.
2. $y^{-1}x \in H$.
3. $y^{-1}xH = H$.

Proof.
$(1 \implies 2)$ Given $xH = yH$ then $\{xh \mid h \in H\} = \{yh \mid h \in H\}$, which means that for some $h_1, h_2 \in H$, we have $xh_1 = yh_2 \implies y^{-1}x = h_2h_1^{-1} \in H$.

$(2 \implies 1)$ Given $y^{-1}x \in H$, then $\forall\, h_1\ \exists\, h_2 \in H$ such that $y^{-1}xh_1 = h_2 \implies xh_1 = yh_2 \implies xH = yH$.

$(1 \implies 2 \implies 3)$ Given $xH = yH$ then $y^{-1}x \in H$. Now consider $y^{-1}xH = \{y^{-1}xh \mid h \in H\}$, but $y^{-1}x \in H$ so $y^{-1}xh \in H$ and thus, by closure of groups, $\{y^{-1}xh \mid h \in H\} = H$.

$(3 \implies 2 \implies 1)$ Given $y^{-1}xH = H$, then $y^{-1}x$ must be in H otherwise group closure of H is broken. Since $y^{-1}x \in H$, we know $xH = yH$. This ends our proof. □

Lemma 1.21. *The left cosets of a subgroup H in G partition the group.*

Proof. To prove that the cosets of a subgroup H in G partition the group, we attempt to find an equivalence relation on H. For $a, b \in H$, we define $a \sim b \iff a^{-1}b \in G$. From the previous lemma, this implies two similar elements will have the same cosets. Hence, we have the equivalence classes as the cosets themselves. Hence, we see the cosets partition the subgroup. □

Definition 1.22. The set of all distinct left cosets of H in G is denoted by G/H, the order of this set is denoted by $[G : H]$, which is called the *index* of H in G.

Lemma 1.23. *All cosets of a subgroup H in G have the same order as H.*

Proof. Let aH be a coset of H in G for $a \in G$. If we define a map $\varphi : H \to aH$ such that $\varphi(h) = ah \; \forall \; h \in H$. Then it is a bijective map, hence the order of H and aH would be the same. □

Theorem 1.24 (Lagrange's Theorem)**.** *Let H be a subgroup of a finite group G, then we have $|G| = |H|[G : H]$ and $|H| \mid |G|$.*

Proof. Since we know that all cosets of H in G will have the same order, and they partition the group, then the order of G should be the product of the order of H and the number of distinct cosets i.e. the index of H in G. Thus, we have the following:

$$|G| = |H|[G : H] \implies |H| \mid |G|$$

□

Corollary 1.25. *If the order of an element is $\mathcal{O}(a)$ for $a \in G$, and $n = |G|$, then $\mathcal{O}(a) \mid n$.*

Proof. Let $H < G$ be a subgroup such that $H = \langle a \rangle$. Then $|H| = \mathcal{O}(a)$. By Lagrange's theorem, we have $|H| \mid |G| \implies \mathcal{O}(a) \mid |G|$. □

Definition 1.26 (Euler Totient Function)**.** Let $\phi : \mathbb{N} \to \mathbb{N}$ be a function such that $\phi(n)$ is the number of coprimes of n, i.e. $\phi(n) = |U_n|$. For a prime p, we have $\phi(p) = p - 1$. This function is called the *Euler totient function.*

Theorem 1.27 (Euler's Theorem)**.** *If $a, n \in \mathbb{N}$ such that* $\gcd(a, n) = 1$*, then*

$$a^{\phi(n)} = 1 \; (mod \; n) \tag{7}$$

Proof. We know that U_n is the set of $x < n$ such that $\gcd(x, n) = 1$, then $a \pmod n \in U_n$. Let $A = \langle a \rangle < U_n$, then by Lagrange's Theorem, $|U_n| = [U_n : A]|A|$. But we know that the $|A| = \mathcal{O}(a)$, $|U_n| = \phi(n)$ and the fact that the number of distinct cosets of A in U_n is only 1, which is the subgroup A itself. Therefore, you now have $\mathcal{O}(a) = \phi(n)$ and hence $a^{\phi(n)} = 1 \pmod n$. □

Corollary 1.28 (Fermat's Little Theorem)**.** *For any prime p and natural number a, we have:*

$$a^p = a \; (mod \; p) \tag{8}$$

Proof. Using Euler's Theorem, since $\gcd(a, p) = 1$ by virtue of p being prime, $a^{\phi(p)} = 1 \pmod p$, but $\phi(p) = p - 1$, so we have $a^p = p \pmod p$. □

Definition 1.29 (Normal Subgroup)**.** Let N be a group and $N < G$, then N is said to be a normal subgroup, denoted by $N \triangleleft G$, if $gNg^{-1} \subset N \; \forall g \in G$.

Proposition 1.30. *If $N \triangleleft G$, then the following statements are equivalent $\forall \; g, h \in G$:*

1. $gNg^{-1} = N$
2. $gN = Ng$

3. $(gN)(hN) = ghN$

Proof. Given that $N \triangleleft G$, then $gNg^{-1} \subset N$, then $gng^{-1} \in N \; \forall \, n \in N \implies \exists \, m \in N$ such that $m = gng^{-1} \implies mg = gn$. Then, $\{mg \mid m \in N\} = \{gn \mid n \in N\} \implies Ng = gN$. Now, we use this to prove 1.
(2 $\implies$ 1) Since $Ng = gN$, then for some $ng \in Ng \; \exists \; gm \in gN$ such that $ng = gm \implies n = gmg^{-1}$. This implies that $N = gNg^{-1}$. □

(1 $\implies$ 3) Using $N = hNh^{-1}$, we can rewrite $(gN)(hN)$ as $(ghNh^{-1})(hN) = \{ghnh^{-1}hm \mid n, m \in N\} = \{ghn \mid n \in N\} = ghN$. Therefore, $(gN)(hN) = ghN$. □
The other proofs are left as an exercise to the reader.

Theorem 1.31 (Quotient Groups)**.** *Let $H \triangleleft G$, then G/H is a group under the operation $(gH)(kH) = gkH$ for $\forall \, g, k \in G$.*

Proof. From 1.30, we know $\forall \, g, k \in G$ we have $(gH)(kH) = gkH$ for $H \triangleleft G$. The operation is associative since group operation is. Restating, $\forall \, gH, kH \in G/H \implies (gH)(kH) = gkH \in G/H$. The identity of G/H would be H, since $(gH)(H) = (H)(gH) = gH \; \forall \, gH \in G/H$. And $\forall \, gH \in G/H \; \exists \; g^{-1}H \in G/H$ such that $(gH)(g^{-1}H) = (g^{-1}H)(gH) = H$. We have proved associativity, closure, identity and inverse, hence G/H is a group when $H \triangleleft H$. □

Definition 1.32 (Center)**.** Let G be a group, then the *center of G*, $Z(G)$ is defined as the set of elements in G that commute with all other elements.

$$Z(G) = \{g \in G \mid gh = hg \; \forall \, h \in G\}$$

Proposition 1.33. *If $Z(G)$ is the center of a group G, then $Z(G) \triangleleft G$.*

Proof. Let $g_1, g_2 \in Z(G)$, then $g_1h = hg_1$ and $g_2h = hg_2 \; \forall h \in G$. Now, $(g_1g_2^{-1})h = g_1(g_2^{-1}h) = g_1(hg_2^{-1}) = h(g_1g_2^{-1})$ i.e. $(g_1g_2^{-1})h = h(g_1g_2^{-1}) \implies g_1g_2^{-1} \in Z(G)$. By subgroup criterion, $Z(G) < G$. Now, $\forall \, z \in Z(G)$ we have $gzg^{-1} = gg^{-1}z = z \in Z(G)$. Hence, $Z(G) \triangleleft G$. □

2 Homomorphisms

2.1 Homomorphism Basics

We begin delve into trying to connect different groups together by the concept of homomorphisms.

Definition 2.1 (Homomorphism)**.** Let $(G, \cdot)$ and $(H, *)$ be two groups and $\varphi : G \to H$. Then φ is said to be a *homomorphism* if $\varphi(a \cdot b) = \varphi(a) * \varphi(b) \; \forall \, a, b \in G$.

Example 2.2.

1. The trivial homomorphism $e : G \to H$ given by $e(g) = e_H \; \forall \, g \in G$.
2. The identity homomorphism $Id : G \to G$ given by $Id(g) = g \; \forall \, g \in G$.
3. The map $a_n : \mathbb{Z} \to \mathbb{Z}$ defined by $a_n(x) = nx \; \forall \, x \in \mathbb{Z}$.
4. The map $\beta_n : \mathbb{Z} \to \mathbb{Z}_n$ defined by $\beta_n(x) = x \pmod n \; \forall \, x \in \mathbb{Z}$.

5. The determinant $\det : \mathrm{GL}(n, X) \to X^\times$ defined by $det(A) =$ determinant of $A \; \forall \; A \in \mathrm{GL}(n, X)$.

6. The map $T : M_n(\mathbb{C}) \to M_n(\mathbb{C})$ defined by $T(A) = A^T \; \forall \; A \in M_n(\mathbb{C})$.

7. The map $\psi : \mathbb{R} \to \mathbb{C}^\times$ defined by $\psi(x) = e^{ix} \; \forall \; x \in \mathbb{R}$.

8. Given G is a group and $N \triangleleft G$, then the canonical projection $q : G \to G/N$ defined by $q(g) = gN \; \forall \; g \in G$.

Lemma 2.3. *Let $\varphi : G \to H$ be a homomorphism, then $\varphi(e_G) = e_H$ and $\varphi(g^{-1}) = [\varphi(g)]^{-1} \forall \; g \in G$.*

Proof. Given $\varphi : G \to H$ is a homomorphism, then $\varphi(e_G e_G) = \varphi(e_G)\varphi(e_G) = \varphi(e_G)$. Using the cancellation lemma, we get $\varphi(e_G) = e_H$. For the second part, $e_H = \varphi(e_G) = \varphi(hh^{-1}) = \varphi(h)\varphi(h^{-1})$, then $\varphi(h^{-1}) = [\varphi(h)]^{-1}$. □

Definition 2.4. Let $\varphi : G \to H$ be a homomorphism, then:

1. If φ is injective, it is a *monomorphism.*

2. If φ is surjective, it is an *epimorphism.*

3. If φ is bijective, it is an *isomorphism.* Then, we say the two groups G and H are isomorphic or congruent, written as $G \cong H$.

Definition 2.5 (Kernel)**.** Let $\varphi : G \to H$ be a homomorphism, then the set of elements in G that map to the identity in H is called the *kernel* of φ.

$$\ker(\varphi) = \{g \in G \mid \varphi(g) = e_H\}$$

Definition 2.6 (Image)**.** Let $\varphi : G \to H$ be a homomorphism, then set of all images to elements in G is called the image of G or image of φ.

$$\mathrm{Im}(\varphi) = \varphi(G) = \{\varphi(g) \mid g \in G\}$$

Lemma 2.7. *Let $\varphi : G \to H$ be a homomorphism, then* 1. $\ker(\varphi) \triangleleft G$ *and* 2. $\varphi(G) < H$.

Proof.

1. Let $a, b \in \ker(\varphi)$, then we have $\varphi(ab^{-1}) = \varphi(a)[\varphi(b)]^{-1} = e_H[e_H]^{-1} = e_H$, hence $ab^{-1} \in \ker(\varphi)$. By subgroup criterion, we get $\ker(\varphi) < G$. Now, for some $g \in G$ and $a \in \ker(\varphi)$, we get $\varphi(gag^{-1}) = \varphi(g)\varphi(a)[\varphi(g)]^{-1}$, and since $a \in \ker(\varphi)$, we get $\varphi(g)\cancel{\varphi(a)}^{e_H}[\varphi(g)]^{-1} = \varphi(g)[\varphi(g)]^{-1} = e_H$, hence $\varphi(gag^{-1}) \in \ker(\varphi)$. This is true for arbitrary $g \in G$ and $a \in \ker(\varphi)$, hence $\ker(\varphi) \triangleleft G$.

   □

2. Let $a, b \in \varphi(G)$ then $\exists \; g, h \in G$ such that $a = \varphi(g)$ and $b = \varphi(h)$. Then $\varphi(gh^{-1}) = \varphi(g)\varphi(h^{-1}) = \varphi(g)[\varphi(h)]^{-1} = ab^{-1}$. Since $\varphi(gh^{-1}) \in \varphi(G)$, we have $ab^{-1} \in \varphi(G)$. By subgroup criterion, we obtain $\varphi(G) < H$.

   □

Proposition 2.8. *Let $\varphi : G \to H$ be a homomorphism, then the following are equivalent:*

1. *φ is a monomorphism.*
2. *$G \cong \varphi(G)$.*
3. *$\ker(\varphi) = \{e_G\}$.*

Proof.
(1 $\implies$ 2) Let $\psi : G \to \varphi(G)$ defined by $\psi(g) = \varphi(g)$. Then ψ is a homomorphism, since φ is one and it is injective, since φ is and surjective since $\forall\, a \in \varphi(G)\ \exists\, g \in G$ such that $\psi(g) = \varphi(g) = a$. Then ψ is bijective and is an isomorphism, hence $G \cong \varphi(G)$. □

(2 $\implies$ 1) Given $G \cong \varphi(G)$, then $\exists\, \psi : G \to \varphi(G)$, where ψ is a bijection. Notice that for each $g \in G\ \exists\, \varphi(g) \in \varphi(G)$, with g being unique for each $\varphi(g)$. So this must be our bijection, hence $\psi(g) = \varphi(g)$. We proved that $varphi(G) < H$, hence φ may or may not be surjective, but φ will be injective, and hence a monomorphism. □

(1 $\implies$ 3) Since φ is monomorphic, then e_H will have a unique preimage in G. We proved that $\varphi(e_G) = e_H$ for any homomorphism, and the unique preimage condition implies that e_G is the only element such that $\varphi(e_G) = e_H$, hence $\ker(\varphi) = \{e_G\}$. □

(3 $\implies$ 1) Given $\ker(\varphi) = \{e_G\}$, let $g, h \in G$ such that $\varphi(g) = \varphi(h)$, then $\varphi(g)[\varphi(h)]^{-1} = e_H$ then $\varphi(gh^{-1}) = e_H$, but then $gh^{-1} = e_G \implies g = h$. Hence, φ is a monomorphism. □

2.2 Isomorphism Theorems

Theorem 2.9 (First Isomorphism Theorem)**.** *Let G, H be two groups and $\varphi : G \to H$ is a homomorphism, then:*

$$G/\ker(\varphi) \cong \varphi(G)$$

Proof. Because I said so. □

Example 2.10.

1. $\det : \mathrm{GL}(n, X) \to X^\times$, then we obtain $\ker(\det) = \mathrm{SL}(n, X)$ and $\det(\mathrm{GL}(n, X)) = X^\times$, hence:
$$\mathrm{GL}(n, X)/\mathrm{SL}(n, X) \cong X^\times$$
2. $\beta_n : \mathbb{Z} \to \mathbb{Z}_n$ where $\beta_n(x) = x \pmod n\ \forall\, x \in \mathbb{Z}$, then we get $\ker(\beta_n) = n\mathbb{Z}$ and $\beta_n(\mathbb{Z}) = \mathbb{Z}_n$, therefore:
$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$$
3. $\varphi : \mathbb{R} \to \mathbb{C}^\times$ defined by $\varphi(x) = e^{i2\pi x}$, here we uncover that $\ker(\varphi) = \mathbb{Z}$ and $\varphi(\mathbb{R}) = \{z \in \mathbb{C} \mid |z| = 1\} = S^1$, hence we obtain:
$$\mathbb{R}/\mathbb{Z} \cong S^1$$

Theorem 2.11 (Second Isomorphism Theorem)**.** *Let G be a group, with $H < G$ and $N \triangleleft G$, then:*

$$H/(H \cap N) \cong HN/N$$

Proof. Let $\varphi : H \to HN/N$ be a map defined by $\varphi(h) = hN$. This is a homomorphism, and that is easy to verify. The kernel of this homomorphism will be $H \cap N$, and the image will be HN/N. By the first isomorphism theorem, we have:

$$H/(H \cap N) \cong HN/N$$

□

Theorem 2.12 (Third Isomorphism Theorem)**.** *Let G be a group with $H \triangleleft G$ and $N \triangleleft G$, with $H \subset N$. Then:*

$$(G/H)/(N/H) \cong G/N$$

Proof. Let $\varphi : G/H \to G/N$ be a map such that $\varphi(gH) = gN$. This is a homomorphism, which is easy to check. The kernel of this homomorphism is N/H, since $g \in N$ if $\varphi(gH) = N$. The image will be G/N, and hence by the first isomorphism theorem:

$$(G/H)/(N/H) \cong G/N$$

□

Example 2.13.

1. Let m and n be positive integers. Using the second isomorphism theorem, we get:
$$m\mathbb{Z}/(m\mathbb{Z} \cap n\mathbb{Z}) \cong (m\mathbb{Z} + n\mathbb{Z})/n\mathbb{Z}$$
Let $\gcd(n, m) = (n, m)$ and $\operatorname{lcm}(n, m) = [n, m]$, then:
$$m\mathbb{Z}/[m, n]\mathbb{Z} \cong (m, n)\mathbb{Z}/n\mathbb{Z}$$
Equating either sides cardinality, we get $mn = (m, n)[m, n]$.

2. For positive integers ℓ, n and m such that $m \mid \ell$ and $n \mid m$, we know that $\ell\mathbb{Z} \triangleleft n\mathbb{Z}$, $m\mathbb{Z} \triangleleft n\mathbb{Z}$ and $\ell\mathbb{Z} < m\mathbb{Z}$. By the third isomorphism theorem,
$$(n\mathbb{Z}/\ell\mathbb{Z})/(m\mathbb{Z}/\ell\mathbb{Z}) \cong n\mathbb{Z}/m\mathbb{Z}$$

3 Permutation Groups

Definition 3.1 (Permutation Group)**.** Let X be a non empty set. Then the set of permutations (or self bijections) of X is defined by:

$$S(X) \coloneqq \{f : X \to X \mid f \text{ is a bijection}\}$$

forms a group under composition, called the *permutation group* of X. If X is of finite order n, then one can index elements of X as $X = \{1, 2, \ldots n\}$ without loss of generality, and $S(X)$ is denoted as S_n.

Example 3.2.

1. $(S_2, \circ) \cong \mathbb{Z}_2$, where $\circ$ denotes composition of maps.
2. Since each symmetry of D_{2n} induces a permutation of vertices of the polygon, we have $D_{2n} < S_n$ and for $n = 3$, $D_6 \cong S_3$.
3. For any group G, $\mathrm{Aut}(G) < S(G)$, where $\mathrm{Aut}(G)$ is the set of all automorphism (self isomorphisms) from G.

A typical element of S_n is $\sigma : \{1, 2, \dots n\} \to \{1, 2, \dots n\}$, so we often denote such a σ by:

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & n \\ \sigma(1) & \sigma(2) & \dots & \sigma(n) \end{pmatrix}$$

Definition 3.3 (Support of σ)**.** Let $\sigma \in S_n$ be a permutation. Then its *support* is the following set:

$$\mathrm{Supp}(\sigma) := \{1 \leq i \leq n \mid \sigma(i) \neq i\}$$

We simplify writing σ by ommitting elements not in $\mathrm{Supp}(\sigma)$.
The product of permutations, $\sigma_2\sigma_1$ where $\sigma_1, \sigma_2 \in S_n$ is given by:

$$\sigma_2\sigma_1 = \begin{pmatrix} 1 & 2 & \dots & n \\ \sigma_2 \circ \sigma_1(1) & \sigma_2 \circ \sigma_1(2) & \dots & \sigma_2 \circ \sigma_1(n) \end{pmatrix}$$

Proposition 3.4. *Two permutations σ_1 and σ_2 commute if they have disjoint supports.*

Proof. Let σ_1 and σ_2 be permutations in S_n with disjoint supports, i.e.

$$\mathrm{Supp}(\sigma_1) \cap \mathrm{Supp}(\sigma_2) = \emptyset$$

This tells us that if $i \in \mathrm{Supp}(\sigma_2)$, then $\sigma_2\sigma_1(i) = \sigma_2(i) = \sigma_1\sigma_2(i)$ and if $j \in \mathrm{Supp}(\sigma_1)$ then $\sigma_1\sigma_2(j) = \sigma_1(j) = \sigma_2\sigma_1(j)$. Since $\{1, 2, \dots, n\} = \mathrm{Supp}(\sigma_1) \cup \mathrm{Supp}(\sigma_2)$, therefore $\sigma_1\sigma_2 = \sigma_2\sigma_1$. □

Definition 3.5 (k-cycle)**.** A *k-cycle* in S_n is a permutation of the following form:

$$\begin{pmatrix} i_1 & i_2 & \dots & i_k \\ i_2 & i_3 & \dots & i_1 \end{pmatrix}$$

where $1 \leq k \leq n$. A k-cycle is often denoted by $(i_1 i_2 \dots i_k)$. A 2-cycle is called a transposition or inversion. Observe that $i_m = \sigma^{m-1}(i_1)$ and $\mathcal{O}(\sigma) = k$.

Example 3.6.

1. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix} = (132)$
2. $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} = (12)(34)$

Proposition 3.7. *Every k-cycle is a product of no less than $k - 1$ transpositions.*

Proof. Let $\sigma = (i_1 i_2 \dots i_k)$ be k-cycle, then one can observe that

$$(i_1 i_2 \dots i_k) = (i_{k-1} i_k) \dots (i_2 i_3)(i_1 i_2)$$

One can verify this by simply checking, and we have $k-1$ transpositions, hence every k-cycle is a product of no less than $k-1$ transpositions. □

Remark 3.8. While considering a cycle, it is important to indicate the ambient permutation group. For instance, the element of S_3, which sends $1 \mapsto 2$ and $2 \mapsto 1$ and fixes the remaining elements (i.e. the element 3), is denoted by (12), and the element of S_4, which sends $1 \mapsto 2$, $2 \mapsto 1$ and fixes the remaining elements (i.e. the elements 3, 4), is also denoted by (12).

Proposition 3.9. *If $\{\alpha_1, \alpha_2, \dots \alpha_r\}$ is a set of pairwise disjoint cycles in S_n of lengths $n_1, n_2, \dots n_r$, respectively, then the order of the permutation $\sigma = \alpha_1 \alpha_2 \dots \alpha_r$ is equal to* $\operatorname{lcm}(n_1, n_2, \dots n_r)$.

Proof. I'm lazy. Might add it later. Also I really don't enjoy combinatorics. Can we do some pure algebra!!! □

Proposition 3.10. *The number of distinct cycles of length r in S_n is ${}^nP_r/r$.*

Proof. Any ordered set of r elements from $\{1, 2, \dots n\}$ gives a cycle of length r, but a cyclic permutation of elements of a cycle gives the same cycle. Thus r number of arrangements of cycle give the same cycle. Since there are ${}^nP_r$ many arrangements of r elements, the proposition follows. □